

EDUCATION

Heritage High School, Frisco, TX

Class of 2027

GPA 4.00 UW / 5.361 W · Rank 11/481 · ACT 36 · AP Scholar with Distinction

AP exams: 9 taken, a 5 on all 9 — Computer Science A, Statistics, Physics 1, Seminar, English Language & Composition, U.S. History, World History, Psychology, Human Geography

Coursework: AP Calculus BC, AP Physics 2, AP Physics 1, AP Computer Science A, AP English Language, AP English Literature, AP Government, AP Economics, Computer Science III, Mobile App Programming, Spanish 2

HONORS & AWARDS

IEEE-EMBS IECBES 2026 — Research Paper Accepted for Presentation ↗

Dec. 9–11, 2026

Kuala Lumpur, Malaysia · 2026 IEEE-EMBS Conference on Biomedical Engineering and Sciences

Semifinalist, Diamond Challenge 2026 (X-Step) · Finalist, [Texas High School Ideas Challenge](#), Texas A&M (X-Step, May 2026) · 1st, TechLit Bridging the Gap Hackathon ([ShopSmart](#), Aug. 2025) · Innovation Prize, PeddieHacks (of 235 participants) · 3rd, Hack4Humanity deeptanshudevatha.com/projects/competitions

RESEARCH

PAPER 1 · COMPUTATIONAL LINGUISTICS / CLINICAL ML

Predicting Cognitive Function Using Transformer-Derived Speech Representations and Longitudinal Coherence Features ↗

May 2025–Present

First Author · Mentor: Joe Xiao (PhD student)

[medRxiv preprint \(Aug. 2026\)](#) medrxiv.org/content/10.64898/2026.08.04.26359730v1 · [Under peer review, Frontiers in Aging](#)

[Neuroscience — submitted August 04, 2026](#) · [Public code](#) github.com/ddevatha1/dementia-mmse-prediction

QUESTION	Can transcribed speech plus the clinical history already on file forecast a patient’s next-visit MMSE (0–30)? Framed as longitudinal, patient-level supervised regression; most prior speech work is single-visit classification.
DATA	DementiaBank Pitt Corpus (controlled access): 126 patients (69 dementia, 57 control), 2–5 visits → 173 samples .
FEATURES	Designed a <u>53-dimensional feature vector</u> : 12 per-visit features — <u>Sentence-BERT discourse coherence</u> (adjacent-sentence similarity, global coherence, semantic slope, topic drift) plus spaCy lexical/syntactic measures (filler rate, 50-token MATTR, inter-sentence repetition, POS-based fragment ratio, named-entity rate) — with 8 visit-to-visit Δ features, 4 temporal features, 20 PCA components of 384-d SBERT embeddings, and 9 clinical/demographic covariates.
MODEL	LightGBM, chosen for small mixed-type tabular data: RobustScaler → correlation pruning ($ r > 0.8$, clinical variables kept) → permutation-importance selection (top 15) → 3× Gaussian augmentation tagged by parent patient → <u>Optuna TPE</u> over 10 hyperparameters, 200 trials, maximizing mean patient-grouped R^2 .
VALIDATION	<u>Diagnosed patient-identity leakage</u> : folds split by visit, so a model could score by recognizing a person. Rebuilt evaluation as 5-fold GroupKFold on patient ID , locking augmented copies to their parent’s fold and scoring only out-of-fold predictions on original samples ; scores fell once leaked identity was removed.
DEBUGGING	Found why entry MMSE, the expected top predictor, contributed almost nothing: <u>two silent data-join bugs</u> (renamed column; string-vs-integer patient ID) imputed one value for every patient. Fixing them was the largest single gain.
RESULTS	R^2 0.840 ± 0.015 · RMSE 2.75 · MAE 2.02 MMSE points · Pearson r 0.917 . Entry MMSE, baseline diagnosis and Blessed score lead permutation importance.
ABLATION	<u>Controlled feature-source ablation</u> : linguistic-only R^2 0.118 , SBERT-only 0.458 , clinical-only 0.716 , clinical + SBERT 0.759 . Clinical history dominates; SBERT beats handcrafted features and adds complementary signal.
INTEGRITY & LIMITS	<u>Audited the manuscript against the code</u> , removing six unused features. Limits: non-nested selection and tuning (CV may be optimistic); one cohort, no external validation; over-prediction below MMSE 10.

RESEARCH (continued)

PAPER 2 · BIOMEDICAL ENGINEERING / WEARABLE SENSING

A Four-Site Plantar-Pressure Insole: Transducer Error Budget, Feature Equivariance, and Sparse-Versus-Dense Agreement in Overground Walking ↗

Jul. 2026–Present

Accepted for Presentation · IEEE-EMBS IECBES 2026 · Kuala Lumpur · Dec. 9–11, 2026

2nd of 5 authors · ML lead · MIT CSAIL first author; UC Santa Cruz AIEA Lab (Prof. Leilani Gilpin) · deeptanshudevatha.com/research/x-step

FRAMING Asks what the team's existing four-site insole (MET1, MET2, MET5, heel; ESP32, 25 Hz) can legitimately measure — a **measurement instrument, not a clinical classifier**. I did not design the hardware; my work starts from its pressure data. No wearer data; no clinical claim.

ML PIPELINE From those readings I built the analysis pipeline: the feature map over rolling 4 s windows (100 samples × 8 channels) to **59 features** — per-site peak, mean, coefficient of variation and pressure–time integral; window-level cadence, stance ratio, anterior load share, aggregate peak and total PTI — and a **540-parameter** logistic-regression head (≈ 0.23 ms per update).

CORRELATION STUDY Designed the sparse-vs-dense study against an independent 32-cell insole archive (**149 recordings, 15 adults**, 298 foot-takes), comparing the cell nearest each of the insole's four sites with its anatomical cluster. Shape survived (**median r 0.72–0.99**), magnitude did not (**peak-to-peak r 0.43–0.74**); four-site center of pressure tracked anteroposterior (**r 0.81**) but not mediolateral motion (**r 0.01**) — a limit of the existing layout's geometry. Descriptive only.

EVALUATION Quantified subject-identity leakage on a synthetic 24-subject, 9-class cohort: **macro-F1 0.885** subject-grouped vs **0.931** ungrouped (majority baseline 0.040) — a gap larger than the spread across classifiers.

PROJECTS & LEADERSHIP

ShopAI ↗ — *Creator & Sole Developer* · live on the [Apple App Store](#)

2025–Present

One grocery search across nine retailers, plus multi-store trip planning · solo rebuild of a team hackathon prototype · [Architecture write-up](#)

ARCHITECTURE React Native/Expo client over a standalone Express/TypeScript backend on **AWS EC2 behind Nginx**; the client holds no search logic — every search, plan and warm-up check runs server-side through a four-layer, one-direction pipeline.

ADAPTERS Designed a **retailer-adapter abstraction** normalizing **nine chains** (seven with live data) into one schema via four integration strategies: OAuth2 client-credentials REST (Kroger, Harris Teeter), anonymous-session GraphQL (Aldi, Sprouts, Publix via Instacart), headless Playwright (Trader Joe's), server-rendered HTML (Whole Foods). Target excluded (terms of service).

SEARCH PATH Query correction → **concurrent fan-out** racing a shared response-time budget → per-store food/relevance filtering and ranking → explicit ok / error / unavailable / pending status per store. A per-ZIP **state machine** (starting → warming → ready) pre-warms OAuth tokens, cookies and the browser session off the request path.

PLANNER A **deterministic optimizer** exhaustively scores all $2^7 - 1 = 127$ non-empty store subsets, so “cheapest” is exact; cheapest, fastest and fewest-stops are single-field sorts; only “balanced” is weighted multi-criteria. **Zero** LLM calls in search or planning.

PRODUCTION Debugged a Node.js setTimeout overflow in a 30-day location cache that stalled the event loop (≈ 1.5 s → 27 s searches) by capping the sweep interval; shipped by feature-flagging off unfinished meal planning, AI assistant, OCR and barcode scanning.

PROJECTS & LEADERSHIP (continued)

KidCode [↗] — *Founder & Lead Instructor* · Frisco, TX

Jun. 2024–Present

124 students · [3-year program](https://deeptanshudevatha.com/projects/kidcode) deeptanshudevatha.com/projects/kidcode · 10 Frisco neighborhoods · **120-class-day curriculum** · **20 projects** · median 10 per student (max 18)

- CURRICULUM** Designed a three-year, 120-class-day, 20-project (6 + 7 + 7) curriculum: Year 1 Python foundations; Year 2 assert-based unit tests, file I/O, JSON, exception handling, classes and inheritance, libraries; Year 3 spaCy NLP — tokenization, lemmatization, sentence segmentation, POS tagging, NER, Matcher rule patterns, and a rule-based chatbot.
- SPIRAL DESIGN** Earlier projects return at a higher level: Mad Libs becomes Auto Mad Libs with spaCy choosing the blanks; the trivia quiz returns JSON-backed with error handling; Year 2 ends by breaking students' own text-cleaning code, which Year 3 opens by testing against spaCy.
- ASSESSMENT** Wrote 13 curriculum-aligned pre/post assessment pairs (same skills and formats, different code). Paired results, **n = 104**: mean **41.7% → 58.8% (+17.1 pp)**; **99/104** improved, 4 unchanged, 1 declined. Observed change only; no comparison group.
- STUDENT WORK** Students completed **1,214** working programs across the 20 projects, from Python basics in Year 1 to spaCy NLP in Year 3 (all **572** Year 3 programs use spaCy).

X-STEP STARTUP · PRODUCT BUILT ON IEEE-EMBS IECBES 2026 ACCEPTED RESEARCH

X-Step [↗] — *Co-Founder & ML/Mobile Lead*

Aug. 2025–Present

Student startup team · working prototype, no wearer data yet · deeptanshudevatha.com/projects/x-step

- PRODUCT** A plantar-pressure insole for people at risk of diabetic foot problems, built by the team: four FSR402 sensors per foot on an ESP32 at 25 Hz, streaming sequence-numbered 28-byte BLE frames to a React Native (Expo) app with live pressure, trends, alerts and a clinician view.
- MY ROLE** Wrote the React Native/Expo app (BLE client, live pressure heatmap, alerts, trends, guided routine; earlier Flask + LLM health-assistant backend) and trained two CNN prototypes on public data (heatmap postures, wound severity). Designed the ML pipeline that turns the stream into alerts: rolling 4 s windows → 59 pressure and gait features → 540-parameter logistic-regression head, fast enough to run between 40 ms samples.

Ayusha Nilayam [↗] — *Programming & Math Instructor* · Hyderabad, India (remote)

Nov. 2025–Jul. 2026

deeptanshudevatha.com/projects/ayusha-nilayam

- SCOPE** Taught 38 weekly Zoom sessions of Python and math to **11 children** at a children's home, plus two weeks in person.
- CONTENT** Python from `print()` through conditionals, loops and functions to independent projects (calculator, number-guessing game); math from equations and inequalities through the unit circle, and statistics through standard deviation.
- CONSTRAINTS** Adapted to one shared laptop, unreliable internet and a Telugu–English language gap with frequent checks for understanding and student explain-backs.

UT Dallas NLP with Python Workshop — *Participant*

Summer 2025

Dr. Chris Davis · Rule-based text simplifier: dependency parsing, lexical substitution, clause splitting, passive-to-active rewriting.

TECHNICAL SKILLS

- ML & NLP** LightGBM, scikit-learn, Optuna (TPE), PyTorch, PCA, permutation importance, GroupKFold, ablation design; spaCy (tokenization, POS, NER, Matcher, dependency parsing), Sentence-BERT; windowed sensor-feature extraction
- LANGUAGES** Python, TypeScript/JavaScript, Java, SQL
- SYSTEMS** React Native/Expo, Node.js/Express, REST, GraphQL, OAuth2, Playwright, AWS EC2, Nginx